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ARTICLE

Fluorescence probe for mercury(II) based on the aqueous synthesis of CdTe quantum dots stabilized with 2-mercaptoetanosulfonate

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The manipulation of the QDs surface chemistry by adjusting the nanocrystal size, morphology and surface capping ligands plays a crucial role on the selectivity and sensitivity exhibited by the QDs towards a given target analyte. In this work, a novel aqueous synthesis of CdTe QDs capped with a thiol compound containing a sulfonate terminal group (SO_3^-), viz., 2-mercaptoethanesulfonate (MES), was thoroughly investigated seeking the subsequent application as fluorescent probes in chemical analysis. The results obtained by using the prepared CdTe-MES QDs in the determination of distinct metal ions demonstrated a high efficiency of these nanomaterials for monitoring Hg(II) levels. Upon optimization, linear working calibration curves for Hg(II) concentrations of up to 0.5 µmol.L^{-1} were obtained, with a determination coefficient of 0.9984. The detection limit was 0.0095 µmol.L^{-1} and the quantification limit was 0.0324 µmol.L^{-1}. The developed approach, assayed in the determination of Hg(II) in tap water samples, provided analytical results similar to those furnished by the reference method. A comparative evaluation of the accuracy and precision of both procedures, carried out by using the Student's t-test and the Fischer test, demonstrated a good agreement with the tabulated values for a 95% confidence's level.

Introduction

Mercury is one of the most toxic metals and widely distributed in the environment, contaminating air, water and soil, not only due to natural phenomena (volcanic activity) but also as a consequence of human activity. It has a very deleterious impact to the environment mostly due to its ability to accumulate in organisms and up along the food chain.

Mercury levels have scaled up following the industrial revolution of the 19th century due not only to the intensive utilisation in industrial processes, such as in the electrolytic production of chlorine and caustic soda, in electrical appliances (lamps, arc rectifiers, mercury cells), in industrial and control instruments (switches, thermometers, barometers), but also as a consequence of its usage in fungicides, antiseptics, preservatives, pharmaceuticals, electrodes, in dental amalgams, etc. Nowadays, the need to reduce anthropogenic sources of mercury, supported by environmental policies and legislation, has resulted in a significant decrease in mercury industrial uses and emissions.[1] Nevertheless, it remains a serious environmental issue demanding the adoptions of several additional measures to further lessen environmental releases and human exposure, either by supporting the replacement of mercury-containing products by mercury-free alternatives, the reduction of coal-fired electricity generation, implementation of safe handling, use and disposal of mercury-containing products and wastes[2], the development of simpler and more expeditious methodologies for faster contamination monitoring, etc.

According to the "Table of Regulated Drinking Water Contaminants"[3] the maximum mercury level that is allowed in drinking water is 2 µg L^{-1}. The concentration of mercury can be accurately determined in air, water, soil, and biological samples (blood, urine, tissue, hair, breast milk, and breath) by a variety of analytical methods. Most of these methods are total mercury (inorganic plus organic mercury compounds) methods based on wet oxidation followed by a reduction step, but methods also exist for the separate quantification of inorganic mercury compounds and organic mercury compounds. Some analytical methods also require the pre-digestion of the sample prior to the reduction to elemental mercury. In the last 3 years, around 300 articles were published concerning the determination of Hg(II) in environmental samples by using different techniques, such as cold-vapour atomic absorption spectrometry (CVAAS)[4,5], cold-vapour atomic fluorescence spectrometry (CVAFS)[6,7], inductively coupled plasma mass spectrometry (ICP-MS).[8,9] These techniques have high performance capabilities in terms of limit of detection and sensitivity. However, they are not accessible to all laboratories being very expensive and relatively difficult to implement in developing countries. In addition, the use of this equipment requires highly trained and experienced scientists.

Among the optical techniques, the use of nanomaterials, such as quantum dots (QDs), have received great interest due to important advantages features such as simplicity and low cost of implementation, no need for qualified technicians and high sensitivity, which make them valuable alternatives for monitoring

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mercury in waters. In the last few years distinct works reported the determination of mercury(II) in artificial and natural samples by using different kinds of quantum dots: CdS [10,11], CdS/ZnS [12], CdTe[13] and Mn:CdS/ZnS. [14]

The choice of QDs as photoluminescent probes for analytical purposes has been explored due to their peculiar combination of optical, electronic and chemical properties.[15]. Despite the numerous advantages afforded by the use of these nanoprobes, some shortcomings have been recognized in both fundamental research and analytical applications. From an analytical point of view, the high photoluminescence (PL) and reactivity are the most important characteristic of QDs, but at the same time this high reactivity could result in limited selectivity towards the target analyte impairing their use as effective fluorescent probes.[16, 17] However, this can be compensated through the manipulation of the QDs surface chemistry by adjusting the nanocrystal size, morphology and surface capping ligands, thus determining their reactivity, quantum yield, stability and solubility in a given solvent.[18, 19] Indeed, the capping ligand plays a crucial role in the selectivity and sensitivity of the QDs to a target analyte.[17] The reactivity and the charge provided by the capping ligand should be considered for any analytical application, which are determined by their terminal functional groups pointing to the outside environment.[17] In the case of CdTe QDs the most commonly used capping ligands are thiol compounds containing carboxylic acid as terminal functional groups, such as thioglycolic acid (TGA) and mercaptopropionic acid (MPA) and also amines namely, L-cysteine (CYS) and glutathione (GSH).[20] Nevertheless, the list of available possibilities for novel cappings in the preparation of aqueous nanocrystals increases continuously.

Aiming to obtain the required chemical functionality and providing surface groups essential for further application of the nanoparticles, in the present work, and to our knowledge for the first time, the aqueous synthesis of CdTe QDs capped with a thiol compound containing a sulfonate terminal group (SO_3^-), viz., 2-mercaptoethanesulfonate (MES), was thoroughly investigated. CdTe nanoparticles were prepared in aqueous solution through the reaction of $CdCl_2$ and NaHTe, as a source for tellurium, in the presence of MES. It should be mentioned that CdSe QDs capped with MES were already reported but these were prepared by an organometallic synthesis and subsequently surface-modified in aqueous media by ligand exchange.[21]

In this work was also investigated the analytical potential of the MES-capped CdTe QDs as fluorescent probes for the determination of distinct metal ions, in order to appraise the sensitivity and selectivity provided by nanocrystals with surface ligands containing thiol and sulfonate groups.

Experimental

Chemical and materials

All solutions were prepared with water from a Milli-Q system (Millipore Inc., Bedford, MA) with resistivity of 18 MΩ cm and all chemicals were of analytical grade and used as received.

A 174 mg L^{-1} Hg (II) stock solution was prepared from HgCl2 purchased from Riedel-deHaen (Germany). Sodium hydroxide (NaOH, 97.0%) purchased from Aldrich (St. Louis, MO, USA) and hydrochloric acid (HCl, 36 %) obtained from Panreac (Barcelona, Spain) were used to prepare buffer solution. Methanol obtained from Sigma-Aldrich (St. Louis, MO, USA) was used to prepare the CdTe quantum dots solution.

The tris(hydroxymethyl)aminomethane-maleate buffer solutions were prepared by mixing 50 mL solution 0.2 mol.L^{-1} tris acid maleate with a 0.2 mol.L^{-1} NaOH solution until to achieve the desired pH (6.0 to 9.0). The resulting solution was diluted to 200 mL with water. The tris acid maleate solution was prepared dissolving 24.2 g tris(hydroxymethyl)aminomethane and 23.2 g maleic acid both purchased from Sigma-Aldrich (St. Louis, MO, USA) in 1000 mL.

The tris-HCl buffer solution was prepared by mixing 50 mL solution 0.2 mol.L^{-1} tris(hydroxymethyl)aminomethane (Sigma-Aldrich, St. Louis, MO, USA) with a 0.2 mol.L^{-1} HCl solution until to achieve the pH 9.0. The resulting solution was diluted to 200 mL with water. The 0.2 mol.L^{-1} tris solution was prepared dissolving 24.2 g tris(hydroxymethyl)aminomethane in 1000 mL water.

The carbonate-bicarbonate buffer solution at pH 10 was prepared with 27.5 mL of 0.2 mol.L^{-1} anhydrous sodium carbonate and 22.5 mL of 0.2 mol.L^{-1} sodium bicarbonate solution purchased from Sigma-Aldrich (St. Louis, MO, USA).

A 1 mol.L^{-1} sodium hydroxide stock solution was prepared by dissolving 40.0 g of NaOH in 1000 mL water.

Tap water samples were collected and spiked with different concentrations of Hg(II).

Apparatus

The fluorescence spectra were obtained at room temperature with a Perkin Elmer LS 50B luminescence spectrometer and a FP-6500 fluorescence spectrophotometer (Jasco, USA). The absorbance spectrum was obtained with a double-beam V-660 spectrophotometer (Jasco). A pH-meter PHS-3C (Zhengji) was used to pH measurements. The morphology, crystalline structure and size distribution of MES-capped CdTe QDs were investigated by high resolution transmission electron microscopy (HRTEM) using a JEOL JEM 3010 microscope (LNNano-CNPEM) operated at 300 kV. An inductively coupled plasma mass spectrometer (model iCAP™ Q ICP MS, Thermo Fisher Scientific, Bremen, Germany) equipped with a MicroMist nebulizer, a baffled cyclonic spray chamber, a standard quartz torch and a two-cone design was used for Hg (II) determination in water for comparison purpose. The ICP-MS instrumental operating conditions used in the Hg (II) determination were as follows: RF power— 1550 W; plasma gas flow—14 L.min^{-1}; auxiliary gas flow—0.8 L.min^{-1}; nebulizer flow rate—0.95 L.min^{-1}.

Synthesis and purification of MES-capped CdTe quantum dots

For the synthesis of MES-capped CdTe quantum dots, tellurium powder (200 mesh, 99.8%), sodium borohydride ($NaBH_4$, 99%), cadmium chloride hemi(pentahydrate) ($CdCl_2.2.5H_2O$, 99%), sodium 2-mercaptoethanesulfonate (MES, ≥ 98%) were purchased from Sigma-Aldrich (St. Louis, MO, USA); absolute ethanol (99.5%) was obtained Panreac (Barcelona, Spain).

The synthesis protocol followed the two-steps method proposed by Zou et al[22] for MPA-capped CdTe nanocrystals, which was modified and optimized for MES, as discussed in the next section. The first step consisted on the reduction of tellurium with $NaBH_4$. Briefly, by using a molar ratio 2:1, tellurium powder and sodium borohydride were placed in a 25 mL two-necked flask fitted with a septum (under N2 atmosphere). Then, 10.0 mL of N_2 saturated distilled water was added by using a syringe. The mixture

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was heated, under stirring, for 30 min at 80 °C, under a slow nitrogen flow, to obtain a deep red NaHTe solution. At the same time, into a three-necked flask, 8×10^{-4} mol of $CdCl_2$ solution and 1.8×10^{-3} mol of MES solution were mixed in a 40 mL N_2 saturated solution. The pH value of the solution was adjusted to 11.5 using 1.0 mol.L^{-1} NaOH solution. Subsequently, in a second stage, the freshly prepared NaHTe solution was transferred, under vigorous stirring, into the Cd precursor solution at room temperature. The addition of the NaHTe solutions was accompanied by a development of an orange color. The mixture was heated to reflux, in oil bath, at 100 °C, under atmospheric conditions. The molar ratio of Cd^{2+}:Te^{2-}:MES was fixed at 1:0.03:2.3. The particle size of the CdTe nanocrystals was controlled by changing the refluxing time.

During the synthesis of CdTe nanocrystals, the temporal evolution of PL spectra was recorded continuously, to control the growth of the particle and size distribution.

The purification of CdTe QDs was achieved by precipitation with absolute ethanol to remove the contaminants, the excess stabilizer and cadmium ions. The precipitate was subsequently separated by centrifugation at 4000 rpm for 10 min, dried under vacuum, kept in amber flasks and protected from light, for posterior use.

Seeking to comparatively evaluate the analytical behaviour of the prepared MES-capped quantum dots with respect to CdTe QDs passivated with other capping ligands, several aqueous colloidal solutions of CdTe nanocrystals passivated with thioglycolic acid (TGA), mercaptopropionic acid (MPA), L-cysteine (CYS) and glutathione (GSH) were prepared according to the literature methods[22-26], with some modifications.

Fluorescence measurement

Into Falcon tubes, 2.0 mL of QDs solution (1 µmol.L^{-1}) and different amounts of Hg(II) solution were sequentially added and the final volume (5 mL) was completed with a Tris-HCl buffer solution (0.1 mol.L^{-1}, pH 9). After 5 min, the fluorescence emission spectra (420-650 nm) were recorded by fixing the excitation wavelength at 400 nm. The slit widths of the excitation and emission were 5 nm and 10 nm, respectively.

Samples were collected and spiked with different Hg(II) concentrations and immediately analysed, in triplicate, according to the proposed method and the standard method (ICP-MS).

Reference procedure

In order to validate the results obtained by the developed methodology, the tap water samples spiked with Hg(II) were also analysed by ICP-MS according to USEPA method 200.8.[27] In all blanks, calibration standards and water samples spiked with Hg(II), 100 µg L^{-1} of gold stock standard was added to avoid memory interference effects.

Results and discussion

Optimization of the hydrothermal synthesis of MES-capped CdTe QDs

In literature, MES has already been used as capping ligand for the synthesis of quantum dots[21], more precisely CdSe QDs. However, in the referred work, the nanocrystals were obtained by using the organometallic synthetic route, which is laborious and troublesome involving the utilization of high-boiling-point pyrophoric organic solvents for capping of the QDs with an initial organic layer of trioctylphosphine/trioctylphosphine oxide (TOP/TOPO). By ligand exchange this organic layer was subsequently replaced by the hydrophilic MES capping, in order to assure solubility in aqueous media. In a different perspective, one of the aims of the present work was to carry out the direct aqueous synthesis of CdTe QDs stabilized with MES. With this purpose, the two-step aqueous synthetic route proposed by Zou et al[22] and using MPA as capping ligand, was adapted and optimized for the preparation of MES-capped CdTe QDs. This optimization involved the evaluation of the impact of several parameters on nanocrystals growth rate and optical properties. The studied parameters were precursors concentration, precursors (Cd:Te:MES) molar ratio and pH.

Precursors concentration was assessed by adjusting pH to 9.5 and the molar ratio (Cd:Te:MES) to 1:0.05:2.3, while the cadmium concentration varied between 5 and 40 mmol.L^{-1}. The obtained results demonstrated that the Cd concentration had no influence on the nanoparticles growth rate but a relevant effect on the optical properties of the obtained QDs. In fact, with the increase of the Cd concentration from 5 to 20 mmol.L^{-1}, the PL intensity of the obtained nanocrystals increased about 1000%. For Cd concentrations higher than 20 mmol.L^{-1}, the PL intensity tended to decrease.

To investigate the influence of Cd:Te molar ratio distinct combinations (i.e. 1:0.03, 1:0.05, 1:0.1 and 1:0.2) were tested while Cd:MES molar ratio was kept at 1:2.3. It was observed that Cd:Te stoichiometric ratio used in the synthesis affected both the nanocrystals growth rate and the PL intensity, which decreased as Te concentration increased. It has been reported[24] that CdTe QDs with low PL intensity exhibited much more Te atoms at the surface than those with high photoluminescence efficiency. Moreover, a partial coverage of surface Te sites by thiol ligands during the synthesis, with the formation of a Cd-SR layer, favours the obtaining of highly luminescent nanocrystals.

Concerning the influence of Cd:MES molar ratio it was verified that for values lower than 1:2.3, the QDs PL intensity was low, increasing for higher ratios. However, the FWHM also increased and a white precipitate deposited at the bottom of the three-necked flask during the refluxing process. This way, for all subsequent assays the molar ratio of Cd:MES was fixed at 1:2.3.

Finally, the study of the pH effect was carried by using the previously optimized conditions while the pH of the solution was changed from 7.5 to 8.5, 9.5, 10.5 and 11.5. The obtained results, depicted in Fig. 1, revealed that the higher the pH the higher the nanoparticle growth rate. Moreover, the highest fluorescence intensity was observed for pH = 11.5.

Characterization of CdTe-MES quantum dots

Absorption and fluorescence spectra of the as-prepared MES-capped CdTe QDs were obtained in order to verify the optical properties and the quantum-confinement of the nanocrystals. The Fig. 2 shows the temporal evolution of both normalized absorption (a) and photoluminescence (b) spectra of the MES-capped CdTe colloidal solutions with different sizes wherein it was possible to observe that all nanoparticles presented good optical properties, such as wide absorption bands and narrow full width at half maximum (FWHM from 46 to 71 nm) of emission peak, confirming that the MES-capped CdTe QDs were nearly monodisperse and homogeneous.

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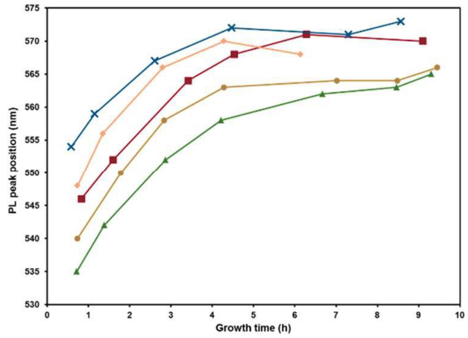

Fig.1 Influence of pH on the nanocrystals growth. Variation of PL maximum emission wavelength for pH values of 7.5 (▲), 8.5 (●), 9.5 (■), 10.5 (♦) and 11.5 (×).

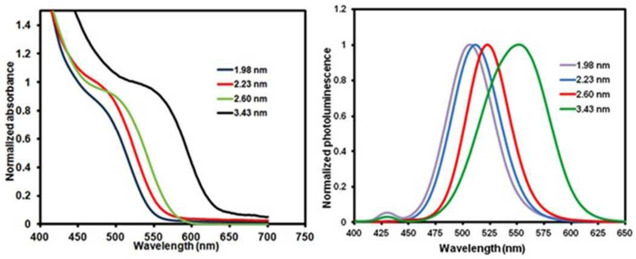

Fig. 2 Normalized absorption and photoluminescence spectra of different sizes of MES-capped CdTe QDs.

About 7 minutes after the beginning of synthesis, the colloid solution emitted a weak green fluorescence, under UV light. With increasing reflux time, the well-resolved absorption maximum of the first electronic transition shifted to longer wavelengths, from 488 to 570 nm, because of the clearly growth of the nanoparticles. At the same time, as the particle size increased the fluorescence emission peak also exhibited a red-shift, due to the quantum confinement effect.

The particle size of the synthesized QDs was calculated according to the Eq.(1) proposed by Yu et al.[28]:

$$D = (9.8127 \times 10^{-7})\ \lambda^3 - (1.7147 \times 10^{-3})\ \lambda^2 + (1.0064)\ \lambda - 194.84 \qquad (1)$$

where D (nm) is the diameter and λ (nm) is the wavelength of maximum absorbance corresponding to the first excitonic absorption peak of the nanocrystal. The estimated diameters of the synthesized nanoparticles were 1.98, 2.23, 2.60 and 3.43 nm.

To facilitate the posterior preparation of the QDs solutions the molar concentrations of the different nanocrystals were calculated.

This was carried out by establishing firstly the molar absorptivity (ε) by using the following equation (Eq.(2)) [28]:

$$\varepsilon = 3450\Delta E(D)^{2.4} \qquad (2)$$

where ΔE is the transition energy corresponding to the first absorption peak and the unit is in eV. Knowing the molar absorptivity (ε), it was simple to reach the molar concentration by measuring the absorbance of a known mass concentration solution of QDs and by applying the Beer-Lambert's law.

With the aim of obtaining relevant information about the morphology, size and structure of the synthesized QDs, the nanoparticles characterization by using high-resolution transmission electron microscopy (HR-TEM) was performed. The HR-TEM images depicted in Fig. 3 showed that the as-prepared CdTe QDs stabilized with MES were quasi-spherical with a good monodispersity. The average size of the MES-capped CdTe QDs was 3 ± 1 nm (Fig. S1 - Supplementary Material). Additionally, well-resolved lattices planes were observed indicating thus that the as-prepared nanoparticles possessed a highly crystalline structure.

Interaction of CdTe quantum dots with metal ions

The QDs reactivity, in particular the sensitivity and selectivity, is determined mostly by the nanocrystals size and the nature of the capping ligand used for their stabilization in aqueous media. Anticipating the use of the QDs as fluorometric probes and to investigate the modulation effect of several metal ions on the QDs photoluminescence properties, different capping ligands (MPA, CYS,

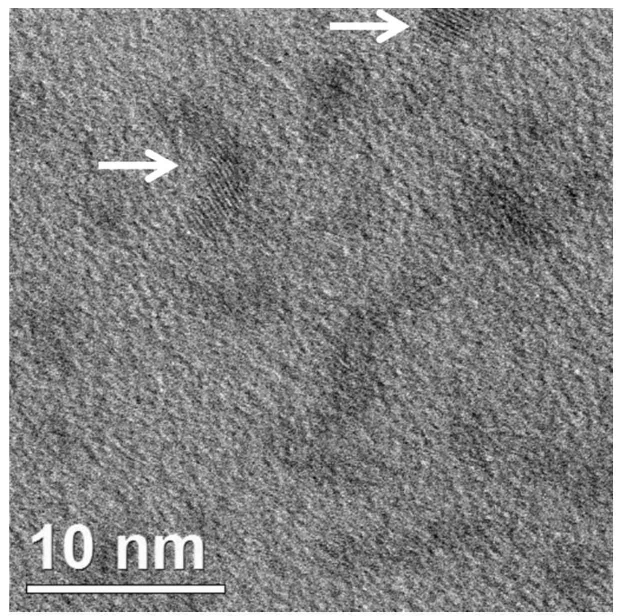

Fig. 3 HR-TEM image of the MES-capped CdTe QD.

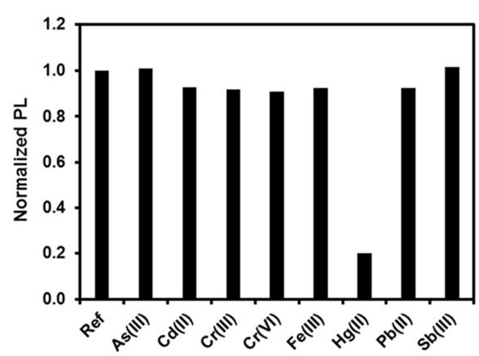

Fig. 4 Influence of several metals ions (2.50 $\mu mol.L^{-1}$) on the photoluminescence intensity of MES–capped CdTe QDs.

GSH and MES) were assayed. In these assays the diameter and molar concentration of the nanocrystals were kept alike (ϕ≈2 nm

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and [QDs] ≈1.00 μmol.L^{-1}) and the metal ions concentration was fixed at 2.50 μmol.L^{-1}. A microplate reader with fluorometric detection (excitation and emission filters of 360/40 and 528/20 nm, respectively) was used to screen the interaction of As (III), Cd (II), Cr (III), Cr (VI), Fe (III), Hg (II), Pb (II) and Sb (III) with each of the different thiol-capped CdTe QDs. The obtained results showed that the QDs passivated with CYS and MES had a higher affinity for Hg (II) (Fig. S2 - Supplementary Material), specially the MES-capped ones, as it could be perceived by the high quenching effect attained, much more pronounced than the one observed for the other tested metals (Fig. 4).

The chelation of mercury by sulfonate-based compounds has been widely reported [29-31] demonstrating the strong affinity between Hg(II) ions and SO_3^- functional groups. Most likely, the Hg(II) ions bind onto the surface of QDs through the functional sulfonate groups of MES, facilitating the electron transfer process between QDs and Hg(II). [32] The close interactions between electron donor-acceptor sites promote an inhibition of the QDs fluorescence intensity due to the electron-hole recombination annihilation on the excited semiconductor nanoparticles. [33]

These results demonstrated that MES-capped CdTe QDs have a noteworthy analytical potential for application in the fluorometric determination of Hg (II) ions. As a consequence, the succeeding assays were intended to optimize the reaction conditions between MES-capped CdTe QDs and Hg (II) ions.

Optimization of the chemical parameters

The optimization of all parameters affecting the determination of mercury (II) was carried with the purpose of reaching the highest sensitivity enabling the analysis of samples with low concentration, such as water samples, and was aimed at the maximization of the PL quenching. This was calculated as F_0/F, representing F_d (blank reading) and F (sample reading) the analytical signal obtained in the absence and presence of Hg(II), respectively.

Table 1 Values of K_{sv} for Hg(II) using different diameter of CdTe-MES (1 μmol.L^{-1}).

MES (nm)	K_{sv} (L.μmol^{-1})
1.98	2.226
2.23	7.422
2.60	4.112
3.43	6.018

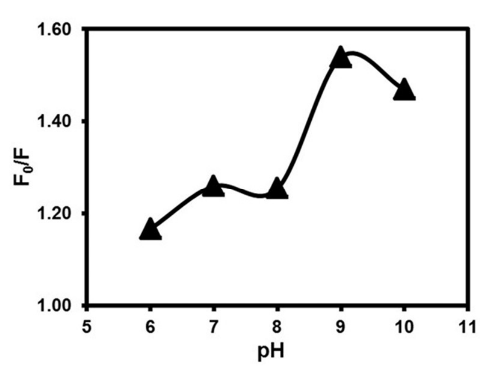

Fig. 5 Effect of the pH on the fluorescence quenching of CdTe-MES QDs upon the interaction with Hg (II).

The chemical parameters influencing the F_0/F were investigated and the optimum values were later used for the analytical determination of Hg(II) in spiked tap water samples. The studied chemical parameters involved the size and concentrations of QDs, solvent, pH, buffer composition, ionic strength, and reaction time.

Effect of QDs diameter and concentration

Bearing in mind that the nanoparticle size can affect the QDs reactivity, the influence of the QDs diameter on the PL quenching induced by Hg (II) was assessed in terms of the Stern-Volmer (Ksv) constant. Four CdTe-MES QDs of different sizes, namely 1.98, 2.23, 2.60, and 3.43 nm were tested, by fixing the respective concentration at 1 μmol.L^{-1}. For each QDs size, calibration curves for Hg (II) concentrations ranging between 0.025 and 0.125 μmol.L^{-1}, were obtained. Each calibration standard was prepared by adding 2 mL MES-CdTe QDs and Hg(II) solution (0.25 μmol.L^{-1}) in a range from 0.5 to 2.5 mL. The final volume (5 mL) was completed with deionized water and the emission spectra were recorded from 420 to 650 nm, fixing the PL excitation wavelength at 400 nm.

The results demonstrated (Table 1) that the higher quenching efficiency (higher Ksv) was obtained when using QDs with a diameter of 2.23 nm, which were therefore selected for the subsequent experiments.

QDs reactivity is also dependent from concentration, which not only influences the photoluminescence intensity (PL) and sensitivity but also the linear working range for Hg (II) determination. CdTe-MES QDs solutions at concentrations ranging from 0.25 to 2.0 μmol.L^{-1} were assayed with Hg (II) 0.125 μmol.L^{-1} of Hg (II) being the photoluminescence intensity measured before and after reaction. The obtained results showed that by increasing the QDs concentrations up to 1.0 μmol.L^{-1}, the quenching effect of Hg (II) on the QDs PL increased, which was related with increased F_0 (blank reading). For QDs concentrations higher than 1.0 μmol.L^{-1} the quenching markedly decreased, clearly because a too high QDs concentration surpasses the analyte quenching effect. Therefore, 1.0 μmol.L^{-1}solutions of MES-capped CdTe QDs were selected for further assays.

Effect of pH, composition and concentration of buffer solution

The reactivity and stability of the QDs in a given solvent are closely related to the surface charge of the nanoparticles which is dependent on the ligand chemical nature and the medium pH. In this sense, the influence of pH on the QDs PL quenching (F_0/F) induced with Hg (II) was investigated from 6.0 to 10.0, using tris-maleate (pH=6.0–9.0) or carbonate (pH=10) buffer solutions. For each pH tested, PL intensity was monitored in the absence and presence of 0.250 μmol.L^{-1}Hg (II) and the observed F_0/F was plotted as a function of the pH value. Fig. 5 clearly shows that the quenching effect of Hg (II) increased with pH from 6.0 to 9.0 and then tended to decrease.

The QDs surface charge or chemistry can be also affected by the buffer composition, which can induce QDs colloidal instability leading to nanoparticles aggregation and consequent PL inhibition.[34] In this regard, three buffers, at pH=9 and at a concentration of 0.2 mol.L^{-1}, but with different chemical composition, namely, tris maleate acid/NaOH (TAM, tris (hydroxymethyl) aminomethane plus maleic acid plus NaOH), tris-HCl (tris-HCl, tris (hydroxymethyl) aminomethane plus HCl) and H_3BO_3-Borax (ABB) were tested. The results revealed that tris-HCl buffer allowed obtaining the higher QDs stability and the higher quenching effect by Hg (II).

Along with pH and buffer composition, the magnitude of the analyte interaction with the QDs surface can be influenced by the ionic strength (buffer concentration).[35] Aiming to evaluate this

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parameter, solutions of MES-capped QDs 1.0 µmol.L^{-1} were made to react with 0.125 µmol.L^{-1}Hg(II) prepared in different concentrations of tris-HCl buffer (ranging from 0.05 to 0.8 mol.L^{-1}). The maximum quenching effect was observed for tris-HCl concentrations of 0.1 and 0.2 mol.L^{-1}, although a better sensitivity (Ksv) was obtained when using a buffer concentration of 0.1 mol.L^{-1}.

In accordance with the preceding results, a tris-HCl buffer solution at pH=9 and at a concentration of 0.1 mol.L^{-1}, was selected for sample analysis.

Effect of reaction time

As it was previously referred, the stability of the QDs solutions is a key issue as it could affect the reproducibility and accuracy of the successive PL measurements carried out during sample analysis. Effectively, a poor solution stability could result in a PL loss not directly related with the analyte concentration. This aspect is even more important if the analysis demands a high reaction time. In this sense, the photoluminescence of the CdTe-MEs QDs was monitored throughout an extended interval of time upon addition of 0.125 µmol.L^{-1} Hg (II). The obtained results, depicted in Fig. 6, showed that after the addition of Hg (II) ions, the PL intensity decrease up to 4 min, stabilizing for higher reaction times. For that reason and in order to ensure the reproducibility of the measurements, a reaction time of 5 min was selected for sample analysis.

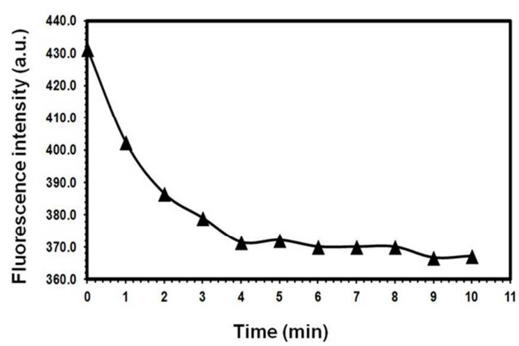

Fig. 6 Effect of the reaction time on the PL intensity of 1 µmol.L^{-1} CdTe-MES QDs upon the addition of 0.125 µmol.L^{-1} Hg(II).

Table 2 Figures of merit for the determination of Hg (II) in water samples using CdTe-MES QDs.

Parameters	Values
Slope (L.µmol^{-1})	2.77
Linear range (µmol.L^{-1})	up to 0.5
R^2	0.9984
RSD (%, n = 6, 0.15 µmol.L^{-1})	3.80
LOD (µmol.L^{-1}, Hg)	0.0095
LOQ (µmol.L^{-1}, Hg)	0.0324

Real sample analysis

By carrying the analysis according to the optimized conditions, a linear relationship between the PL quenching signal (F0/F) and the concentration of Hg(II) was obtained for concentrations values of up to 0.50 µmol.L^{-1}. The linear regression equation was represented by $(F_0/F) = 2.77\ C + 0.9835$, with determination coefficient and relative standard deviation of 0.9984 and 3.80, respectively. The detection limit calculated as following the IUPAC criteria was 9.5 x 10^{-3} µmol.$L^{-1}$of Hg(II) (Table 2).

For the purpose of assessing the accuracy of the proposed method, three natural waters doped with different Hg(II) concentrations were analysed by the proposed fluorometric methodology and the obtained results were compared with those obtained by the reference procedure (Inductively Coupled Plasma with Mass Spectrometry (ICP-MS)) described by US-EPA Method 200.8 (Table 3).[27]

The results obtained by both methodologies were statistically compared using the Student's t-test and the variance ratio F-test. Regarding to the accuracy, a paired Student's t-test confirmed that there were no statistical differences ($t_{calculated}$ = 0.71, $t_{tabulated}$ = 4.30) between the results obtained by the proposed and reference methodologies, for a confidence level of 95% (n = 3). In terms of precision, by applying the variance ratio F-test for a confidence level of 95%, no significant differences between the results obtained by both procedures was verified ($F_{calculated}$ = 2.68, $F_{tabulated}$ = 19).

Considering the obtained recovery values (Table 3), ranging from 92 to 107%, for the determination of Hg(II) in water samples spiked with different concentrations of the analyte, the proposed method showed a good selectivity since the presence of other metals in the samples do not interfere in the detection of Hg(II).

Table 3 Determination of Hg(II) in natural waters using the proposed method and compared with Inductively Coupled Plasma with Mass Spectrometry (ICP-MS)

Sample	Hg(II) Added µmol.L^{-1}	Proposed method µmol.L^{-1}	Recovery (%)	ICP-MS µmol.L^{-1}
1	0.075	0.080 ± 0.002	107	0.070 ± 0.004
2	0.100	0.097 ± 0.003	97	0.113 ± 0.002
3	0.125	0.115 ± 0.001	92	0.124 ± 0.005

Conclusions

In this work, the preparation of water soluble MES-capped CdTe QDs by using a two-pot aqueous synthetic route was for the first time described. This is at the same time a simpler and more environmental friendly alternative to the previously reported organometallic synthetic route used to prepare CdSe-MES QDs.

After optimization of the conditions of synthesis, such as the precursor concentration, Cd:Te:MES molar ratios and pH, highly luminescent, crystalline and monodisperse nanocrystals were obtained. Moreover, a suitable surface functionality was achieved for the sensitive and selective detection of Hg(II) ions. The interaction between Hg(II) and CdTe-MES QDs showed that this metal ion efficiently quenched the photoluminescence intensity of the nanoparticles while for the same concentration, other metals have no noticeable effect on the QDs optical properties. This fact evidenced that the nature of the capping ligand and the pH of the reaction media can have an important role in the selectivity of the proposed methodology.

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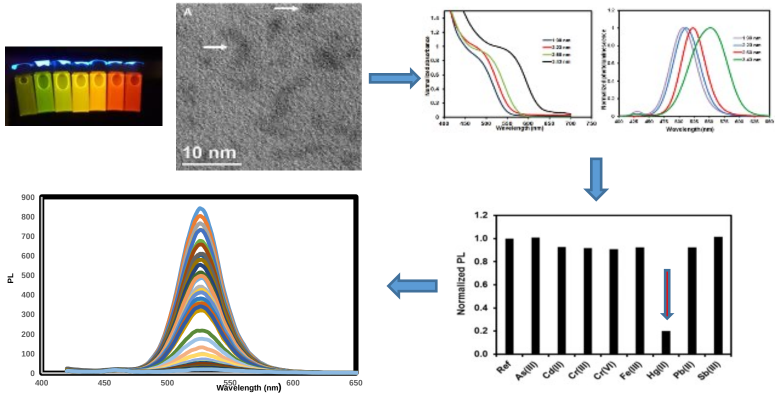

A new capping ligand, 2-mercaptoethanesulfonate (MES), for CdTe quantum dots passivation provide high selectivity and sensitivity in Hg(II) determination